MERAKI - BIOS : RE-NEET 2026
TARGET 360/360 IN BIOLOGY
THREE [3] ALL INDIA FREE TESTS IN BIOLOGY
27.05.2026; 07.06.2026, 17.06.2026

PRACTICE TEST: REVISE AND RECALL: RESPIRATION IN PLANTS

1. Consider the given two statements:

A: During oxidation within a cell, all the energy contained in respiratory substrates is not released free into the cell, or in a single step.

B: The energy released by oxidation in respiration is not or rather cannot be used directly but is used to synthesise ATP.

1. Only A is correct
2. Only B is correct
3. Both A and B are correct
4. Both A and B are incorrect

[PAGE 154 - Recall - Why is ATP called 'energy currency' of the cell?]

2. During glycolysis, when 3-phosphoglyceraldehyde (PGAL) is converted to 1,3-bisphosphoglycerate (BPGA):

1. A molecule of ATP is used
2. A molecule of ATP is produced
3. A molecule of CO_2 is released
4. NADH + H^+ is formed from NAD^+

[Page 156 - Recall - What is the net yield of ATP in glycolysis]

3. In fermentation, by yeast:

A: pyruvic acid is completely oxidised to carbon dioxide and ethanol.

B: pyruvic acid is reduced to lactic acid.

1. Only A is correct
2. Only B is correct
3. Both A and B are correct
4. Both A and B are incorrect

[Page 157 - Recall - what function is accomplished by the fermentation of pyruvic acid]

4. Identify the correct statements regarding aerobic respiration:

A: For aerobic respiration to take place within the mitochondria, the final product of glycolysis, pyruvate is transported from the cytoplasm into the mitochondria.

B: The complete oxidation of pyruvate takes place by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO_2.

C: The passing on of the electrons removed as part of the hydrogen atoms to molecular O_2 results in the simultaneous synthesis of ATP.

1. Only A and B
2. Only A and C
3. Only B and C
4. A, B and C

[Page 158; Recall the fate of pyruvic acid in different conditions]

5. Two successive steps of oxidative decarboxylation during the Tricarboxylic Acid Cycle lead to the formation of:

1. Citric acid
2. Alpha ketoglutaric acid
3. Succinyl-CoA
4. Oxaloacetic acid

[Page 159; Recall - Steps of Krebs / TCA cycle - figure 12.3]

6. The continued oxidation of acetyl-CoA via the TCA cycle requires the continued replenishment of:

A: Oxaloacetic acid

B: NAD^+

C: FAD^+

1. Only A and B
2. Only A and C
3. Only B and C
4. A, B and C

[Page 159; Recall - the summary equation for the complete oxidation of pyruvic acid in aerobic respiration]

7. The mitochondrial electron transport system is present:

1. in the outer mitochondrial membrane
2. in the inner mitochondrial membrane
3. in the intermembrane space
4. in the mitochondrial matrix

[Page 159; Recall - the ultrastructure of mitochondrion from Cell: The Unit of Life]

8. Identify the correct statements regarding the mitochondrial electron transport system:

A: Electrons from NADH produced in the mitochondrial matrix during citric acid cycle are oxidised by an NADH dehydrogenase (complex I), and electrons are then transferred to ubiquinone located within the inner membrane.

B: Ubiquinone also receives reducing equivalents via $FADH_2$ (complex II) that is generated during oxidation of succinate in the citric acid cycle.

C: Cytochrome c is a small protein attached to the inner surface of the outer membrane and acts as a mobile carrier for transfer of electrons between complexes III and IV.

D: Complex IV refers to cytochrome c oxidase complex containing cytochromes a and a3, and two magnesium centres.

1. Only A and B

2. Only A and C

3. Only B and D

4. Only C and D

[Page 160; Recall - the components of the mitochondrial electron transport system]

9. In the mitochondrial electron transport system, Iron-Sulfur (Fe-S) clusters act as essential prosthetic groups that mediate electron transfer. They are not involved in:

1. Complex I

2. Complex II

3. Complex III

4. Complex IV

[Page 160 - Recall - Figure 12.4]

10. Identify the incorrect statement:

1. If fatty acids were to be respired they would first be degraded to acetyl CoA and enter the respiratory pathway.

2. Glycerol would enter the respiratory pathway after being converted to PGAL

3. Proteins and amino acids cannot enter the respiratory pathway

4. Because the respiratory pathway is involved in both anabolism and catabolism, it would hence be better to consider the respiratory pathway as an amphibolic pathway rather than as a catabolic one.

[Page 162 - Recall - amphibolic nature of respiratory pathway]